

Cryptography

Computational Security, PRGs, and Reductions

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August 31, 2026

Announcements

- No class or office hours on Monday, September 7 (Labor Day)
- Homework 1 due Monday, September 14 by 12:45pm on Gradescope
- Project topic suggestions will be posted this week

Lecture 1 Review

- Probability review
- Perfectly-secure private-key encryption
- One-time pad

Outline

- 1 Limitations of Perfect Secrecy
- 2 Computationally-Secure Encryption
- 3 Pseudorandom Generators
- 4 Proofs by Reduction
- 5 Security of PRG+OTP

The One-Time Pad

One-Time Pad Encryption Scheme

- Let $\mathcal{M} = \mathcal{K} = \mathcal{C} = \{0, 1\}^\ell$
- Gen: $k \leftarrow \mathcal{K}$
- Enc: $c = k \oplus m$
- Dec: $m = k \oplus c$

Correctness:

$$\text{Dec}_k(\text{Enc}_k(m)) = k \oplus (k \oplus m) = m$$

Security:

$$\Pr[M = m \mid C = c] = \Pr[M = m]$$

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Take Away

Perfectly secure encryption must have keys as long as the message.

Exercise

Consider OTP encryption:

- What happens if $k = 0^\ell$?

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- What happens if $k = 0^\ell$?
- Modify Gen to never output $k = 0^\ell$. Is this still secure?

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Computational Security

Any PPT adversary breaks security with at most negligible probability.

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 - PPT algorithms calling PPT subroutines remain PPT
 - $\text{poly}(n) \cdot \text{negl}(n) = \text{negl}(n)$
 - Polynomially many negligible risks remain negligible

Redefining Encryption Functionality

Private-key (symmetric-key) encryption scheme:

- Gen, Enc, Dec run in polynomial time
- Gen: $k \leftarrow \text{Gen}(1^n)$ – n is the *security parameter*
- Enc(k, m): $c \leftarrow \text{Enc}_k(m)$ for $m \in \{0, 1\}^*$
- Dec(k, c): $m = \text{Dec}_k(c)$

Correctness

For all n , all k output by $\text{Gen}(1^n)$, and all $m \in \{0, 1\}^*$,

$$\text{Dec}_k(\text{Enc}_k(m)) = m.$$

Redefining Encryption Security

$\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n)$

- $\mathcal{A}$ outputs m_0, m_1 with $|m_0| = |m_1|$
- Challenger chooses $k \leftarrow \text{Gen}(1^n)$ and $b \leftarrow \{0, 1\}$
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Π has *indistinguishable encryptions in the presence of an eavesdropper* if, for all PPT $\mathcal{A}$,

$$\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n).$$

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- Security allows only negligible advantage
- Security must hold for every PPT adversary

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- Recall: $\text{Enc}_k(m) = m \oplus k$
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Idea 2

Stretch k into something longer that looks random.

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Construct deterministic, polynomial-time

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- s must be uniformly random and unknown

PRG Definition

Let $G : \{0, 1\}^n \rightarrow \{0, 1\}^{\ell(n)}$ be deterministic and polynomial-time.

$\text{PRG}_{\mathcal{D}, G}(n)$

- Challenger chooses $b \leftarrow \{0, 1\}$
- If $b = 0$: $r \leftarrow \{0, 1\}^{\ell(n)}$ (random)
- If $b = 1$: $s \leftarrow \{0, 1\}^n$ and $r = G(s)$ (PRG)
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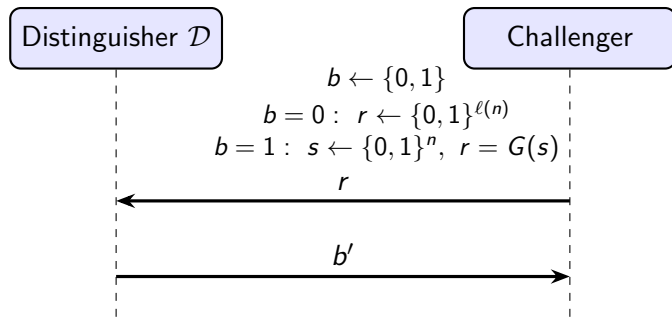
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G is a secure PRG if $\ell(n) > n$ and, for every PPT $\mathcal{D}$,

$$\Pr[\text{PRG}_{\mathcal{D}, G}(n) = 1] \leq \frac{1}{2} + \text{negl}(n).$$

PRG Security Game



Winning Condition

$$b' = b \quad \Pr[\text{PRG}_{\mathcal{D}, G}(n) = 1] \leq \frac{1}{2} + \text{negl}(n)$$

An Example

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- Distinguisher wins with probability

$$\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4} > \frac{1}{2} + \text{negl}(n)$$

Not a secure PRG.

Encryption from a PRG

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After Break

How to prove security of PRG+OTP

10-minute break

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Security proof:

- If the assumption is true, then Π is secure
- Reduction from security of Π to the assumption

Proofs by Reduction: Intuition

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- Show: if Π is insecure, then assumption X is false
- Essentially a proof by contradiction

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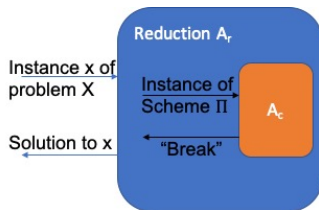
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 - $\mathcal{A}_r$ simulates an instance of Π for $\mathcal{A}_c$
 - $\mathcal{A}_r$ uses $\mathcal{A}_c$ to solve X



Proofs by Reduction: Example

Assumption: $G : \{0, 1\}^n \rightarrow \{0, 1\}^{n+2}$ is a PRG

Goal: Prove $G'(s) = G(s)_{1,\dots,n+1}$ is a PRG

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Fully Black-Box Reductions

- Such reductions are called *fully black-box*
- (Almost) all reductions in cryptography are fully black-box

Outline

- 1 Limitations of Perfect Secrecy
- 2 Computationally-Secure Encryption
- 3 Pseudorandom Generators
- 4 Proofs by Reduction
- 5 Security of PRG+OTP

Security of PRG+OTP: Intuition

Assumption: $G : \{0, 1\}^n \rightarrow \{0, 1\}^{\ell(n)}$ is a PRG

Goal: PRG+OTP is secure against an eavesdropper

PRG+OTP Encryption

- $\text{Gen}(1^n)$: $k \leftarrow \{0, 1\}^n$
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- $\mathcal{A}_r$ runs $\mathcal{A}_c$ and outputs “PRG” if $\mathcal{A}_c$ wins

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PRG+OTP

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PRG Game

$d = 0 : r \leftarrow \{0,1\}^{\ell(n)}$
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Encryption Game

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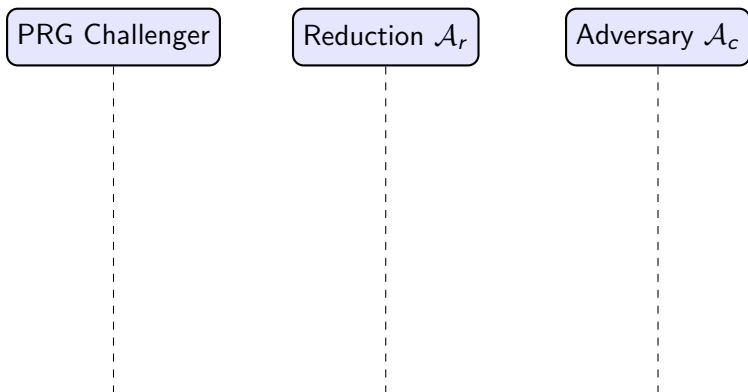
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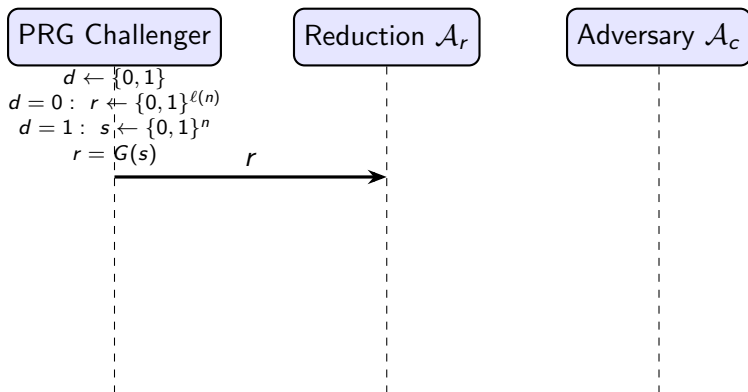
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- Output 1 (“PRG”) if $b = b'$; output 0 otherwise

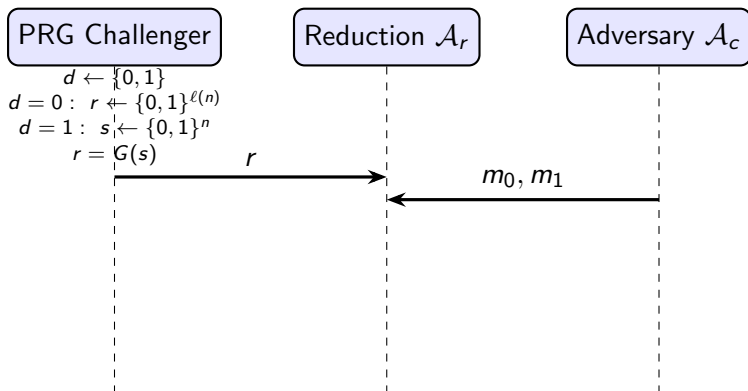
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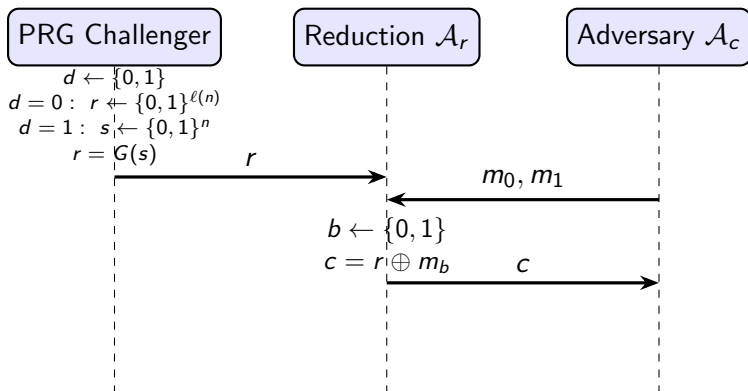
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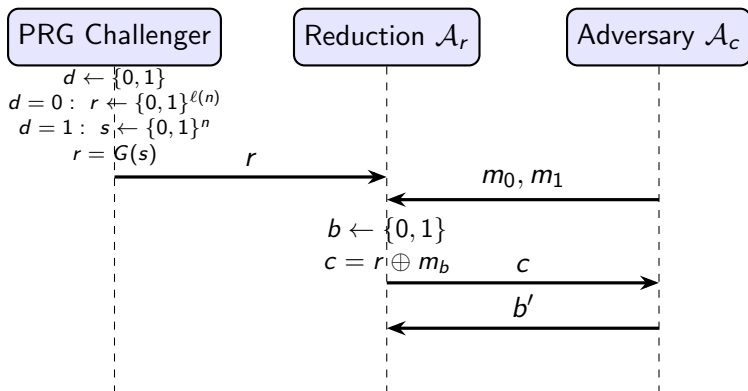
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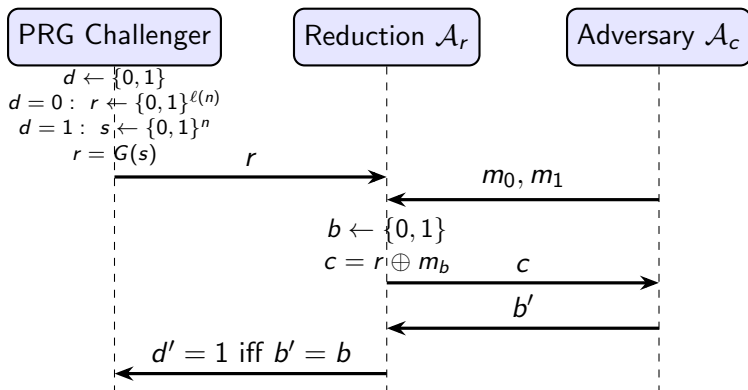
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Therefore,

$$\begin{aligned}\Pr[\text{PRG}_{\mathcal{A}_r, G}(n) = 1] &\geq \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \left(\frac{1}{2} + \frac{1}{\text{poly}(n)} \right) \\ &= \frac{1}{2} + \frac{1}{2\text{poly}(n)}.\end{aligned}$$

Contradiction!

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Next Step

Chosen-plaintext security and randomized encryption